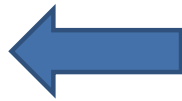


## 3- ARM CORTEX-M ARCHITECTURE

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- **History of ARM**
- **Cortex-M Features**
- **Cortex-M Instruction Set Architecture**
- **Instruction Set**
- **Memory Access**
  - Memory-mapped I/O
- **Exception Handling**



## REFERENCES AND RECOMMENDED READING

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- ARMv7-M Architecture Reference Manual (2014) [ARM DDI 0403](#)
- ARM® Cortex® -M4 Processor Revision: r0p1 (2015) Technical Reference Manual [ARM 100166\\_0001\\_00\\_en](#)
- Cortex™-M4 Devices Generic User Guide (2010) [ARM DUI 0553<sup>a</sup>](#)
- ARM Cortex-M for Beginners: An Overview of the ARM Cortex-M processor family and comparison - Joseph Yiu - [link](#)
- ARM Cortex-M3 Introduction - ARM University Relations - presentation
- ARM Architecture Overview - presentation
- The ARM Architecture (with focus on Cortex-M3) - Joe Bungo, ARM University Program - presentation
- ARM Cortex-M Processor Family - presentation
- The Future is in Your Hands - Peter Middleton - presentation
- How to Choose Your ARM Cortex-M Processor - Tim Menasveta

## 3.1 - ARM HISTORY

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ARM is a British company with focus on microprocessor and GPU design. It currently is owned by the SoftBank Group.

Since Sep,2020 there is an undergoing negotiation with NVidia.

Although ARM was founded in 1990, its story begins in 1981 in the Acorn Computer Ltd.

The success of ARM over the last 3 decades is an inspiring story.

## 3.1 - ARM HISTORY

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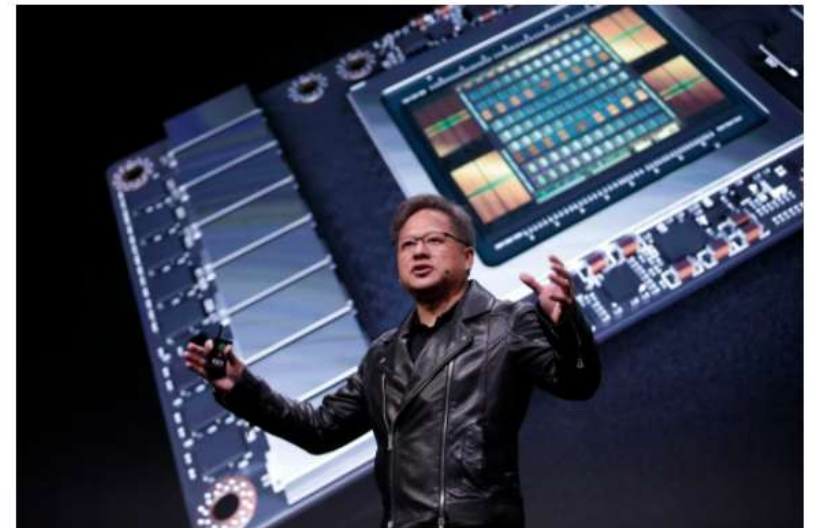
ARM is a British company with focus on mobile processor design. It currently is owned by the SoftBank Group. Since Sep, 2020 there is an ongoing acquisition by Nvidia. Although ARM was founded in 1990, its roots go back to Acorn Computer Ltd.

The success of ARM over the last 3 decades has been remarkable.

The New York Times

### *Nvidia Buys Arm From SoftBank for \$40 Billion*

The move gives Nvidia a way into mobile chips, but the acquisition is likely to face regulatory scrutiny.



Jensen Huang, Nvidia's chief executive, pledged to keep operating Arm as it has been. Ritchie B. Tongco/EPA, via Shutterstock

By Don Clark

Published Sept. 13, 2020 Updated Dec. 1, 2020

SAN FRANCISCO — Nvidia said on Sunday that it would acquire the [British chip designer Arm](#) from SoftBank in a deal worth about \$40 billion.

## 3.1 - ARM HISTORY

---

The BBC micro was released by Acorn Computer Ltd in 1981 as part of the BBC Computer Literacy Project.

The main characteristics of the BBC micro are:

- CPU: 2 MHz 8-bit processor 6502
- Memory: up to 128 KBytes
- Display: PAL/NTSC TV  
640 x 256 pixels (8-bit/pixel)
- I/O: keyboard, joystick, parallel printer, serial



## 3.1 - ARM HISTORY

In 1984, a new set of requirements for the home computer for a replacement of the 8-bit 6502. The Motorola 68000 was considered; however, its interrupt response was not suitable for intended use.

“Steve Furber is one of the brightest guys I’ve ever worked with – brilliant ” says Acorn CEO Hermann Hauser, “and when we decided to do a microprocessor on our own I made two great decisions – I gave them two things which National, Intel and Motorola had never given their design teams: the first was no money; the second was no people. The only way they could do it was to keep it really simple.”

The recent work of Patterson with the Berkeley RISC the MIPS project by Hennessy at Stanford caught the attention of the engineers at Acorn. Hence, they decided to design their own 32-bit RISC processor.

Starting in Oct 1984, in April 1985 the ARM1 was born.

At this time, ARM meant Acorn RISC Machine and was the name of a project at Acorn Computer.

## 3.1 - ARM HISTORY

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ARM2 was released in 1987 and was used in the BBC Archimedes 300, a successor of the BBC Micro.

Characteristics:

- CPU: ARM2
- Memory: up to 16 MBytes



source: [wikimedia.org](https://commons.wikimedia.org/wiki/File:BBC_Archimedes_300.jpg) (CC)

## 3.1 - ARM HISTORY

---

ARM was founded in 1990 with the name Advanced RISC Machines. It had financial support from VLSI, Acorn and Apple.

ARM was created with the purpose of designing processors, thus, becoming one of the first Intellectual Property companies in the world.



ARM Ltd. Headquarters - 1990

source: ARM

## 3.1 - ARM HISTORY

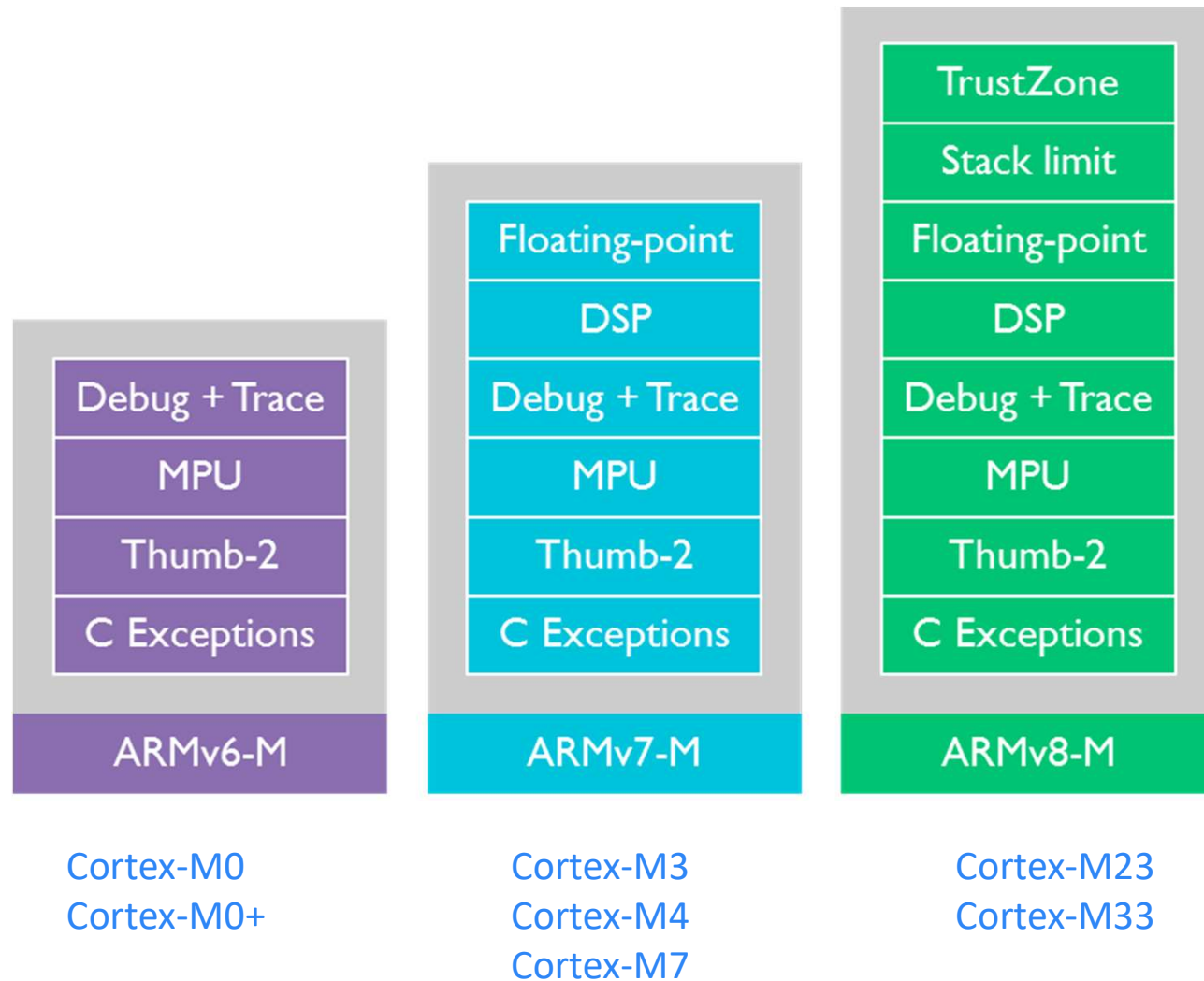
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In 1994 the ARM7TDMI was released, being a major success on the market.

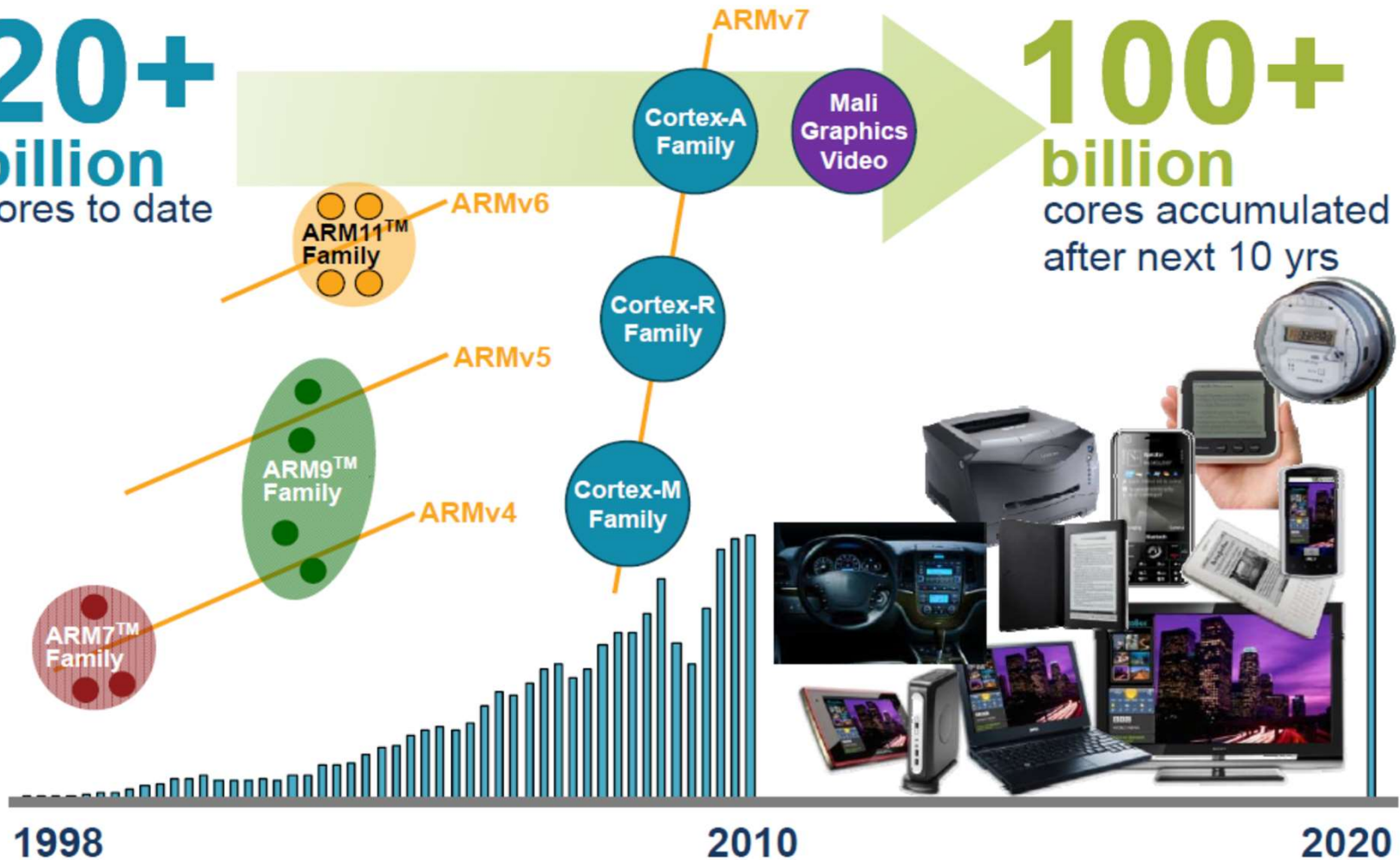
In 2004 the Cortex-M, Cortex-R and Cortex-A series were released.

## 3.1 - ARM HISTORY

Over time, several ARM architecture versions were released. From the seminal ARMv1 to the ARMv8.3-A, released in 2017. The documentation of these standards is available on the [arm.com](http://arm.com) website. For each architectural version there are several implementations.



**20+**  
billion  
cores to date



**100+**  
billion  
cores accumulated  
after next 10 yrs

## Arm-based chip shipments

100bn

(billions)

34%

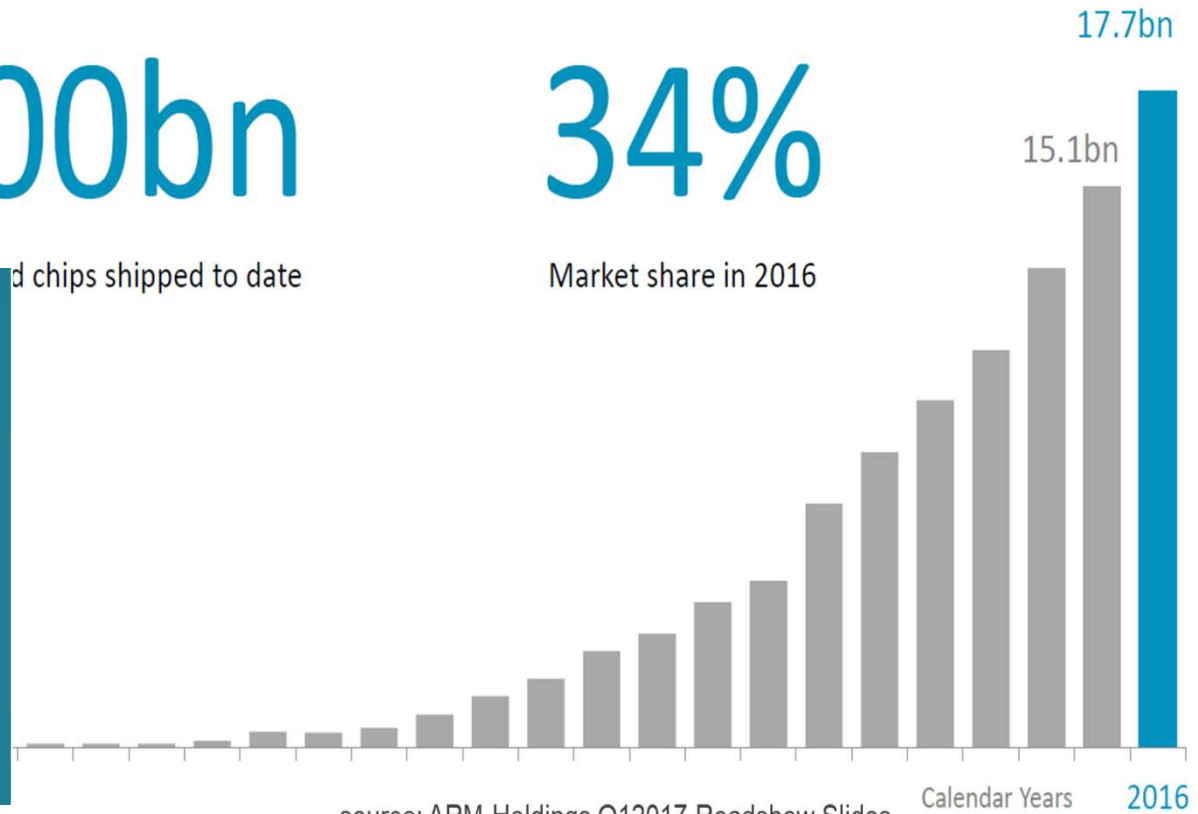
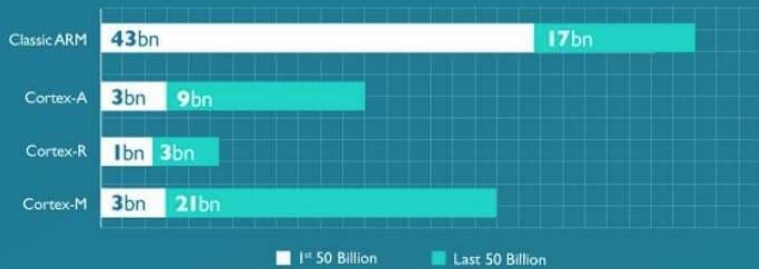
Arm-based chips shipped to date

Market share in 2016

17.7bn

15.1bn

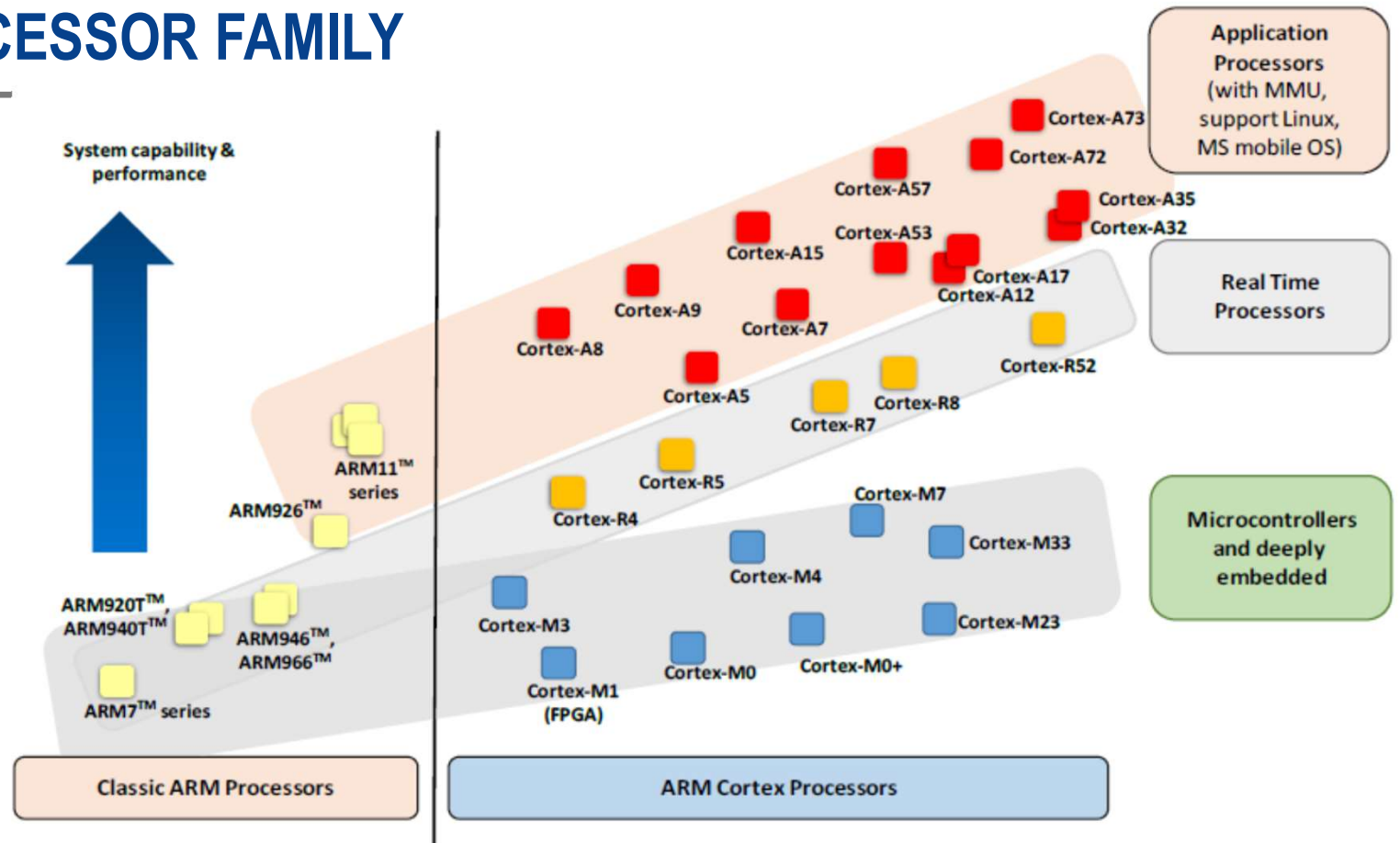
### 1<sup>ST</sup> 100 BILLION CHIPS BY THE NUMBERS PRODUCTS



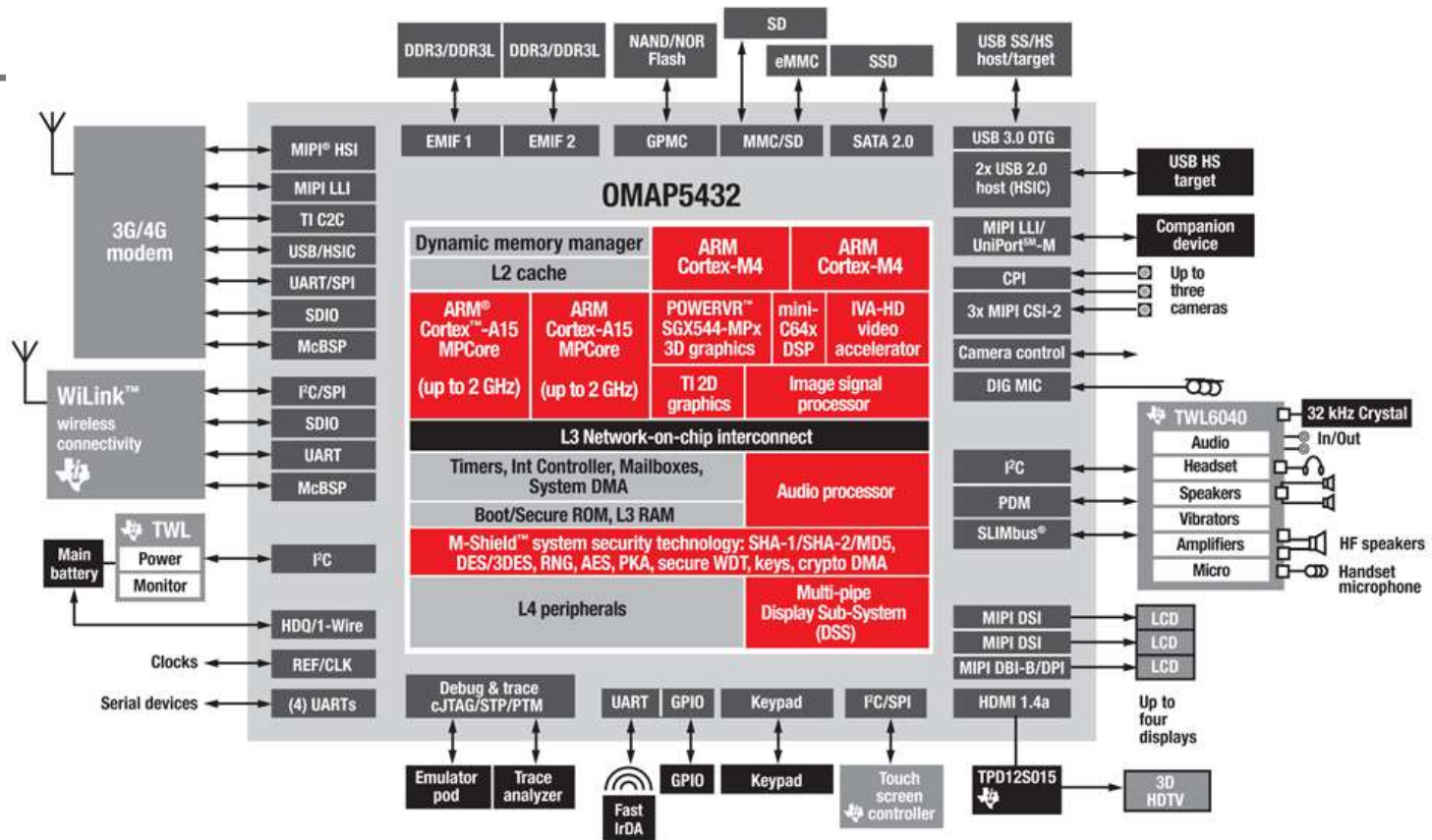
source: ARM Holdings Q12017 Roadshow Slides

source: community.arm.com

# ARM PROCESSOR FAMILY



# TI OMAP5432 SoC

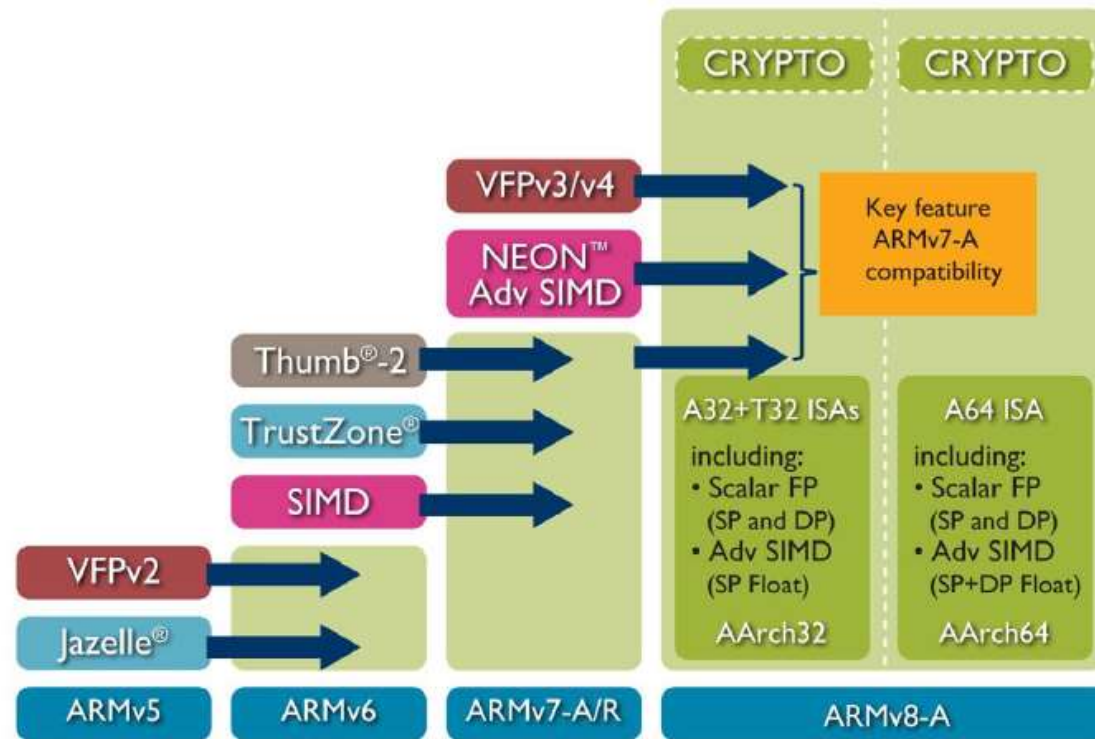


# CORTEX-A FAMILY (32 AND 64-BIT PROCESSORS)

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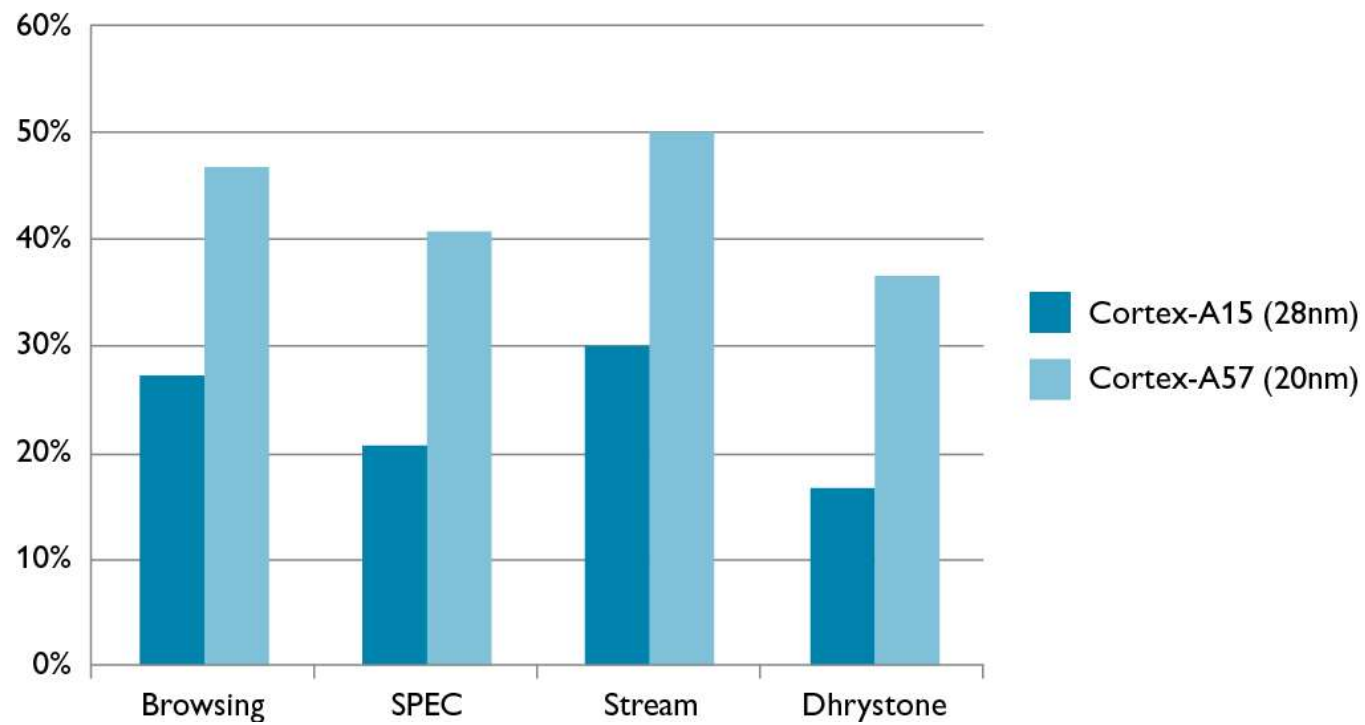
| 64-bit |              | 32-bit |            |
|--------|--------------|--------|------------|
| Year   | Core         | Year   | Core       |
| 2012   | Cortex-A53   | 2005   | Cortex-A8  |
| 2012   | Cortex-A57   | 2007   | Cortex-A9  |
| 2015   | Cortex-A35   | 2009   | Cortex-A5  |
| 2015   | Cortex-A72   | 2010   | Cortex-A15 |
| 2016   | Cortex-A34   | 2011   | Cortex-A7  |
| 2016   | Cortex-A73   | 2013   | Cortex-A12 |
| 2017   | Cortex-A55   | 2014   | Cortex-A17 |
| 2017   | Cortex-A75   | 2016   | Cortex-A32 |
| 2018   | Cortex-A76   |        |            |
| 2018   | Cortex-A65AE |        |            |
| 2018   | Cortex-A76AE |        |            |
| 2019   | Cortex-A77   |        |            |
| 2020   | Cortex-A78   |        |            |

# ARMv8-A (64 BITS)



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### Performance Improvement vs. Cortex-A15 (28nm)



# Cortex™ -A53

ARM CoreSight™ Multicore Debug and Trace

ARMv8 32b/64b CPU  
Virtual 40b PA

NEON™  
SIMD engine with  
cypto ext.

Floating Point  
Unit

8-64k I-Cache  
w/parity

8-64k D-Cache  
w/ECC

Core  
1

2

3

4

ACP

SCU

L2 w/ECC (512kB ~ 2MB)

128-bit AMBA ACE Coherent Bus Interface

# Cortex™-A57

ARM CoreSight™ Multicore Debug and Trace

ARMv8 32b/64b CPU  
Virtual 44b PA

NEON™  
SIMD engine with  
crypto ext.

Floating Point  
Unit

48k I-Cache  
with DED parity

32k D-Cache  
w/ECC

Core  
1

2

3

4

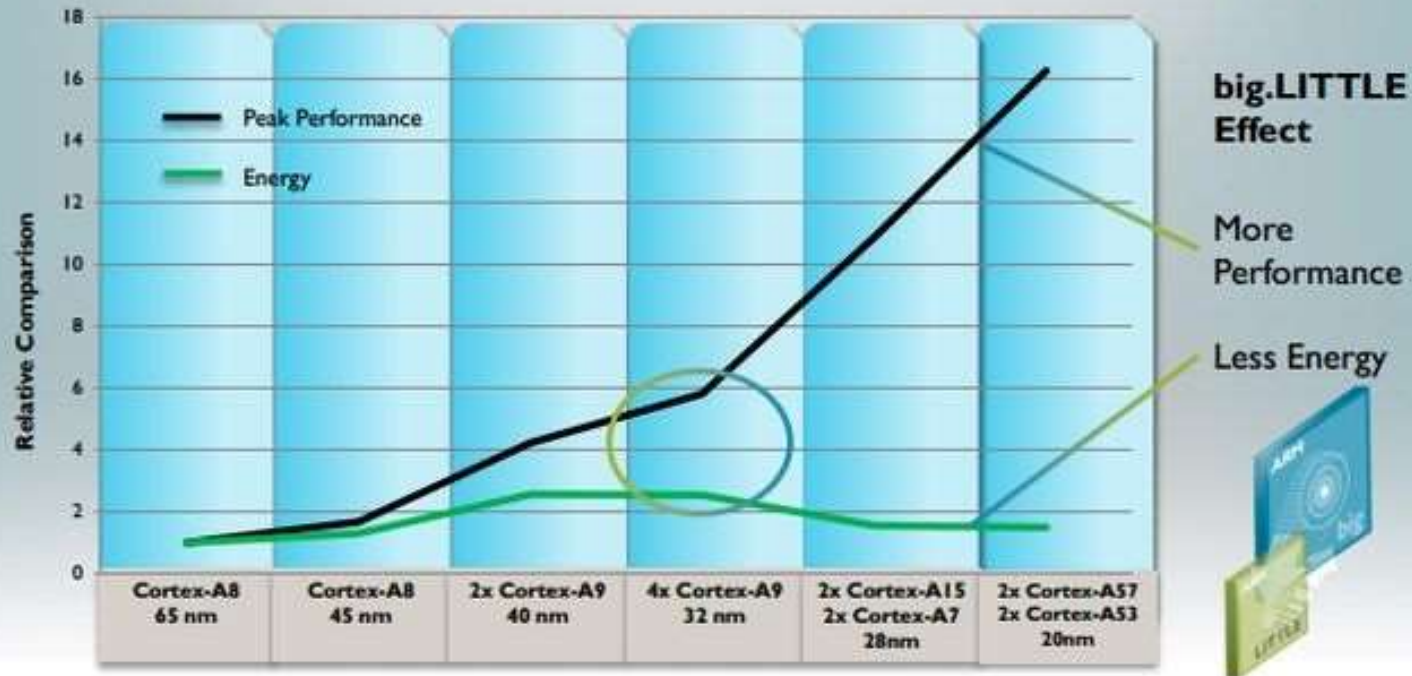
ACP

SCU

L2 Cache w/ECC (512kB ~ 2MB)

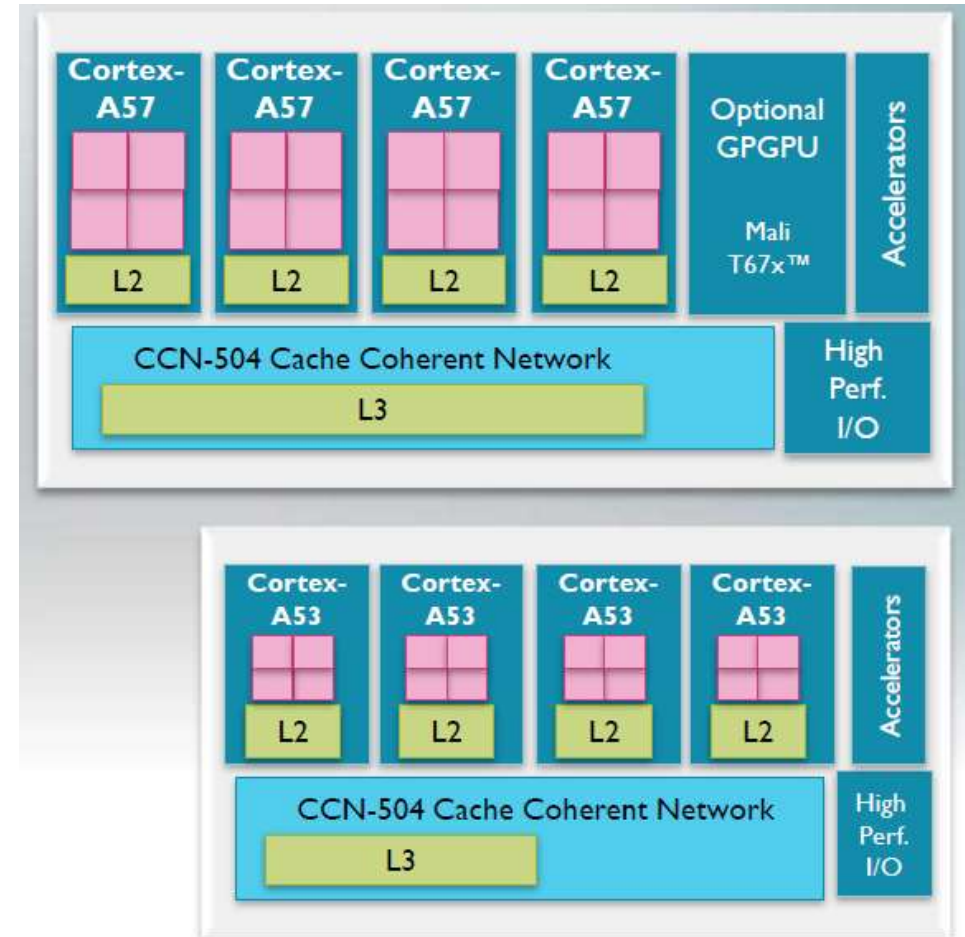
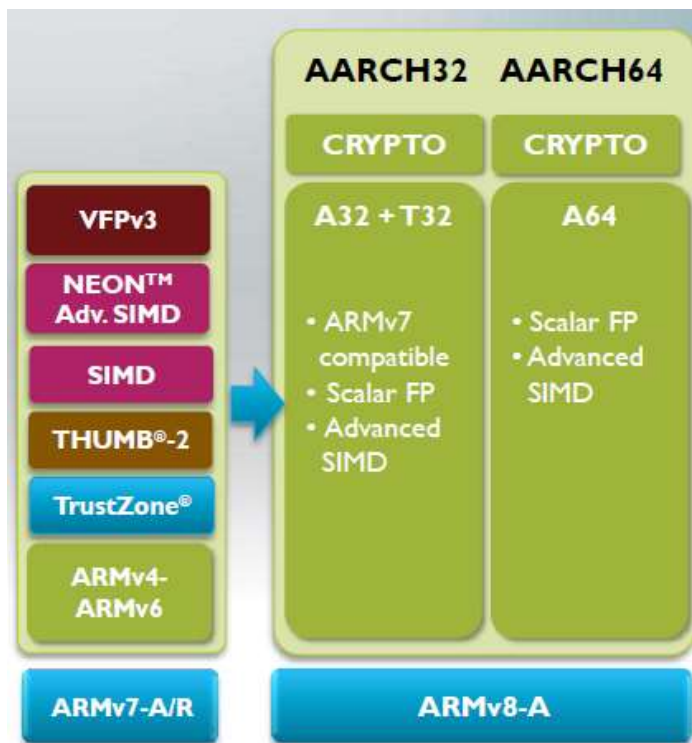
128-bit AMBA ACE Coherent Bus Interface

## Cortex-A50 Series: Faster, Better, Lower Power

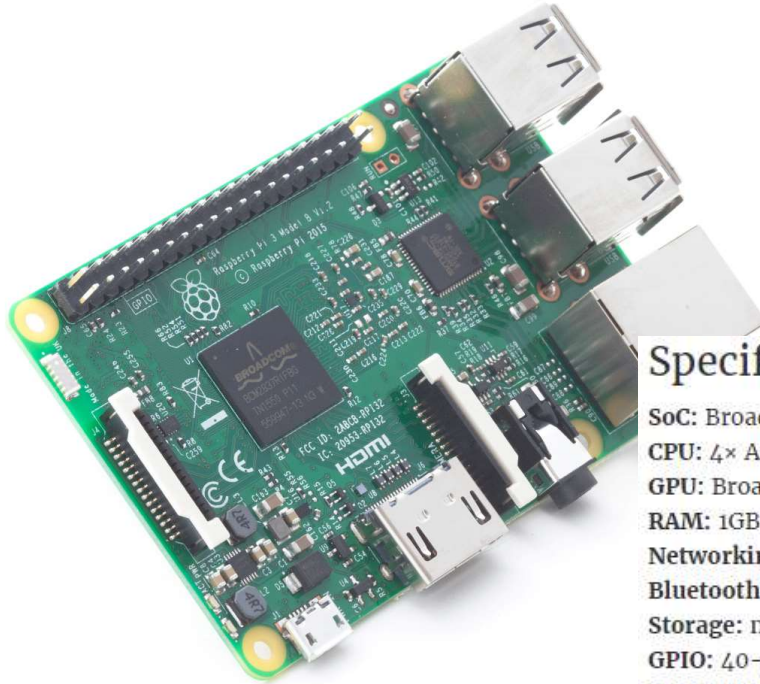


- Continuous improvement on performance and efficiency
- Innovation beyond process technology limitations

# 64-BIT ARM



# FEV 29, 2016 – RASPBERRY PI 3



## Specifications

SoC: Broadcom

CPU: 4× ARM

GPU: Broadcom

RAM: 1GB LPD

Networking: 1

Bluetooth: Blu

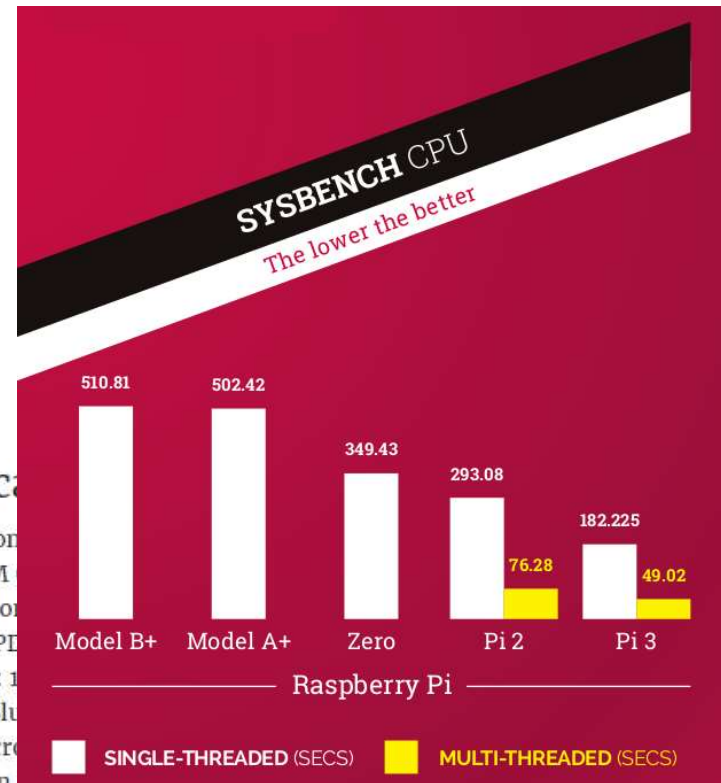
Storage: micro

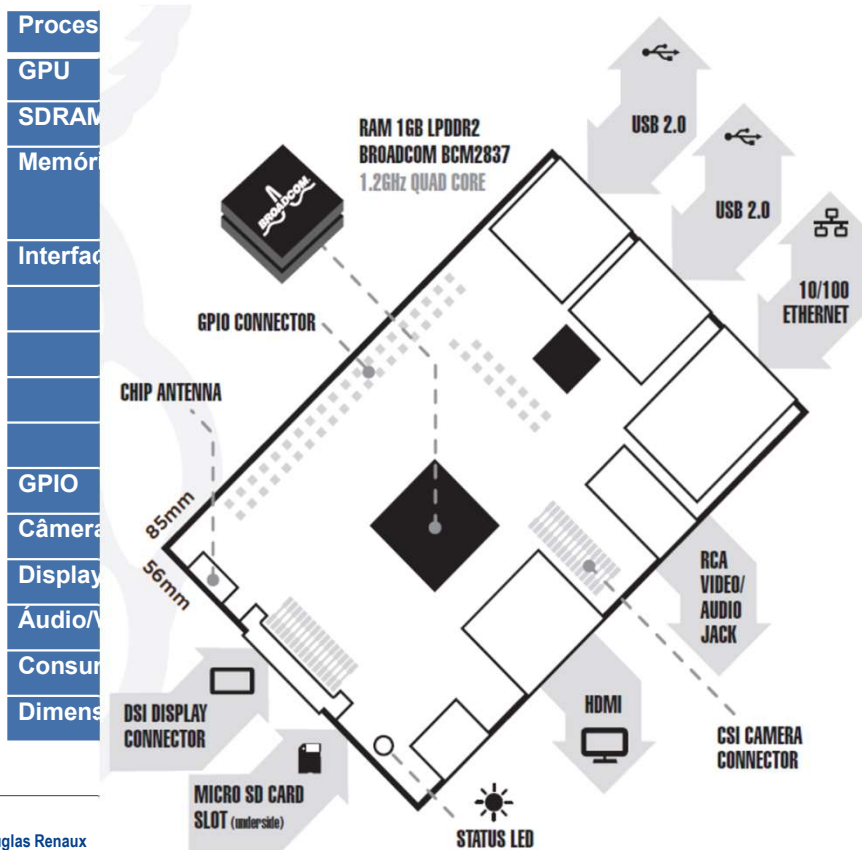
GPIO: 40-pin

Ports: HDMI, 3.5mm analogue audio-video jack, 4× USB 2.0,

Ethernet, Camera Serial Interface (CSI), Display Serial Interface

(DSI)





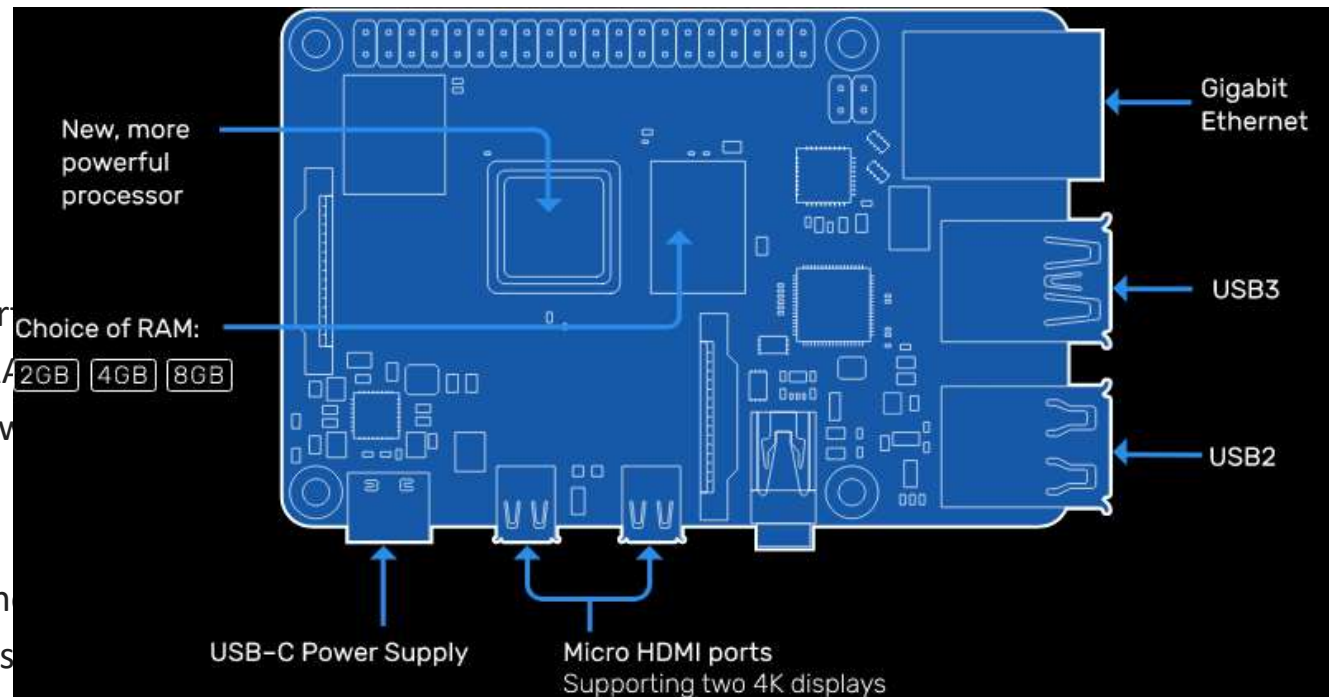
# RASPEBERRY PI 4

June 2019



# RASPEBERRY PI 4

- Broadcom BCM2711, Quad core Cortex-A72 (ARMv8) 64-bit SoC
- 2GB, 4GB or 8GB LPDDR4-3200 SDRAM
- 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless
- Gigabit Ethernet
- 2 USB 3.0 ports; 2 USB 2.0 ports.
- Raspberry Pi standard 40 pin GPIO header
- 2 × micro-HDMI ports (up to 4Kp60 supported)
- 2-lane MIPI DSI display port
- 2-lane MIPI CSI camera port
- 4-pole stereo audio and composite video port
- H.265 (4Kp60 decode), H264 (1080p60 decode, 1080p30 encode)
- OpenGL ES 3.0 graphics
- Micro-SD card slot for loading operating system and data storage



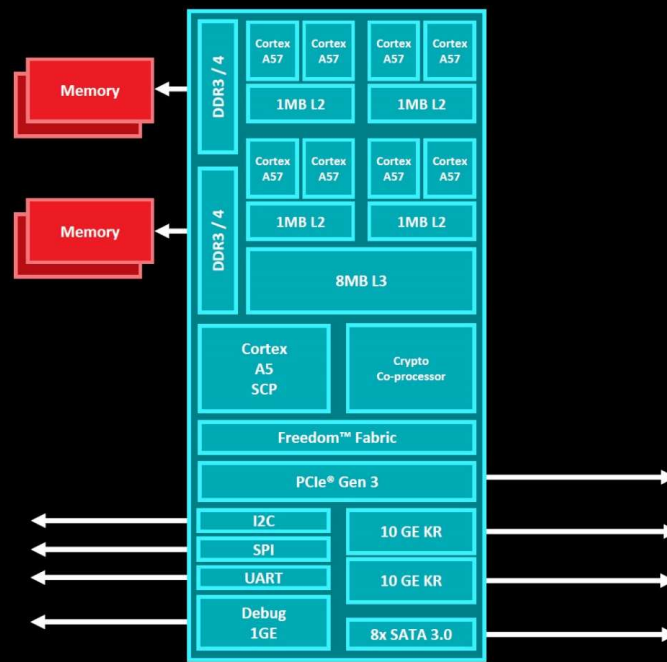
## 64-BITS - AMD

Hierofalcon will p  
up to 2GHz.

In addition to new  
Ethernet and PC  
be a good fit for c

21-Out-2013

### ARCHITECTURAL OVERVIEW OF THE R-SERIES "HIEROFALCON" SOC



- ▲ ARM® Cortex-A57 based architecture
  - Up to eight Cortex™ A57 cores with 4MB total L2 cache
  - 28nm process technology
  - 27mm x 27mm SP1 BGA package
- ▲ Cache Coherent Network
  - Full cache coherency
  - 8MB L3 cache
  - SMMU: I/O address, mapping and protection
- ▲ High Performance Flexible Memory
  - Two 64-bit DDR3/4 channels with ECC up to 1866MHz
  - SO-DIMM, UDIMM, RDIMM support
  - Up to 128GB per CPU
- ▲ Highly Integrated SOC with Flexible I/O
  - Two 10GbE KR
  - Eight SATA 3 (6Gb/s) ports
  - 8 lanes PCIe® Gen 3 (supports 1x8 or 2x4 or 1x4+2x2)
  - SPI, UART, I2C interfaces
- ▲ System Control Processor
  - TrustZone® technology for enhanced security
  - Dedicated system management 1GbE (RGMII)
- ▲ Crypto Co-processor
- ▲ Freedom™ Fabric

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NVIDIA:

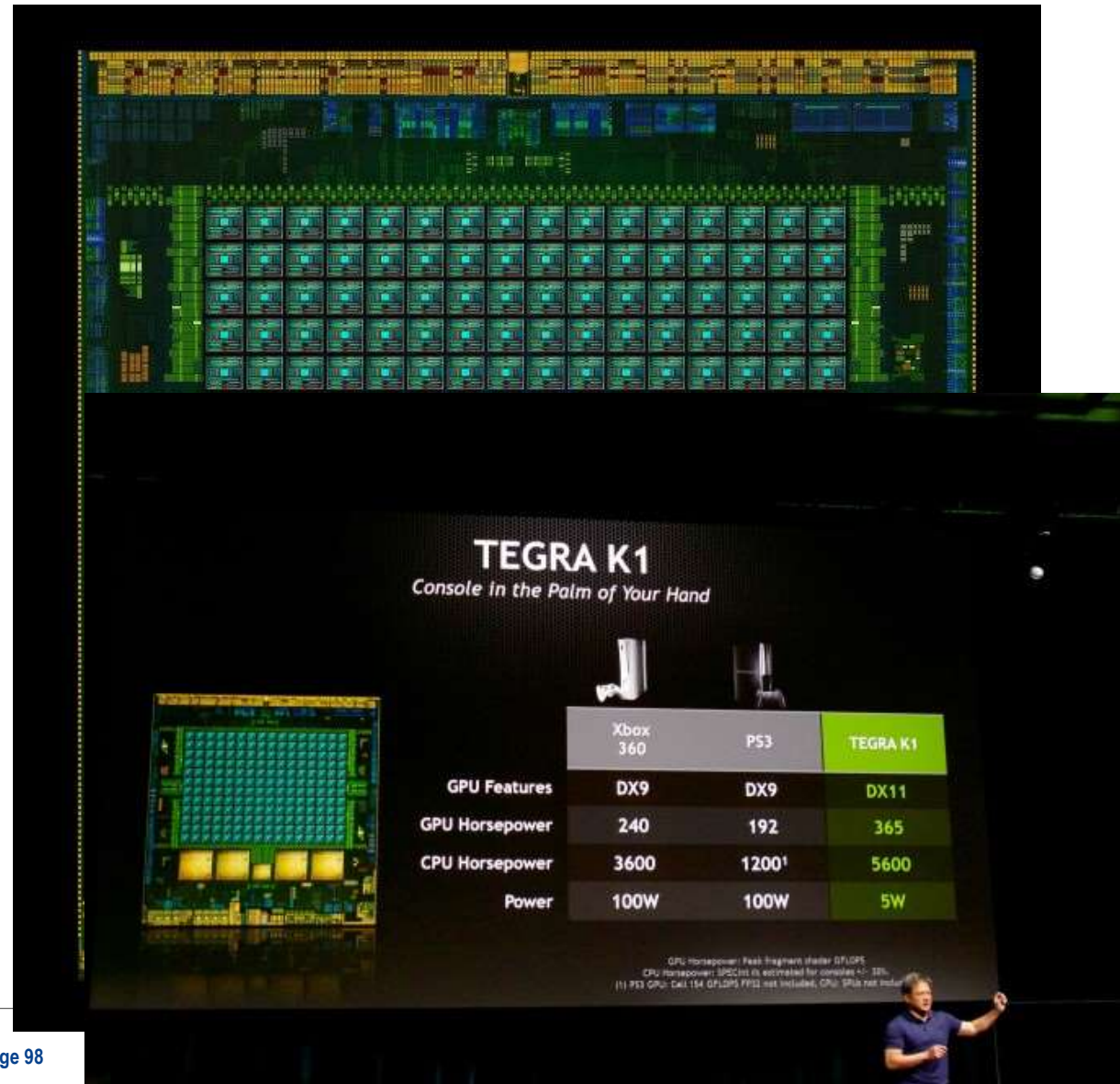
192 GPU

4x A15

ou 2x A57

Tegra K1

Lançado em 2014



**TEGRA K1**  
*Console in the Palm of Your Hand*

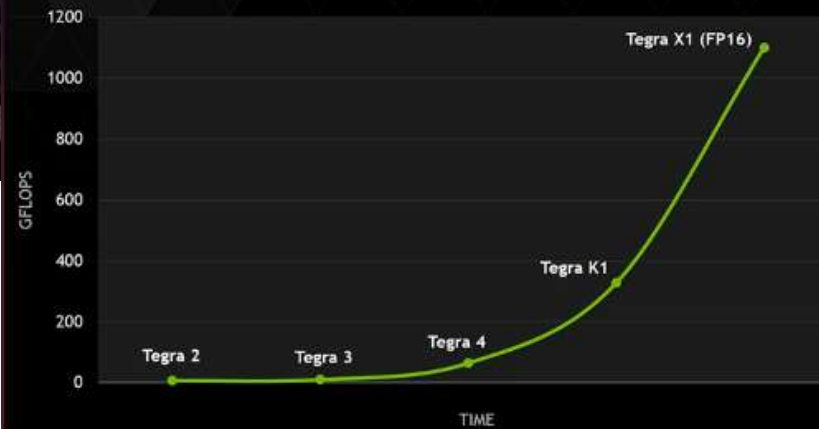
|                | Xbox 360 | PS3                 | TEGRA K1 |
|----------------|----------|---------------------|----------|
| GPU Features   | DX9      | DX9                 | DX11     |
| GPU Horsepower | 240      | 192                 | 365      |
| CPU Horsepower | 3600     | 1200 <sup>(1)</sup> | 5600     |
| Power          | 100W     | 100W                | 5W       |

GPU horsepower: Peak fragment shader GFLOPS.  
CPU horsepower: SPECint64 rate normalized for consoles at 33%.  
(1) PS3 CPU: Cell 154 GFLOPS FP32 not included; CPU, GPU not included.

TEC



## WORLD'S 1<sup>ST</sup> TERAFLUPS MOBILE PROCESSOR

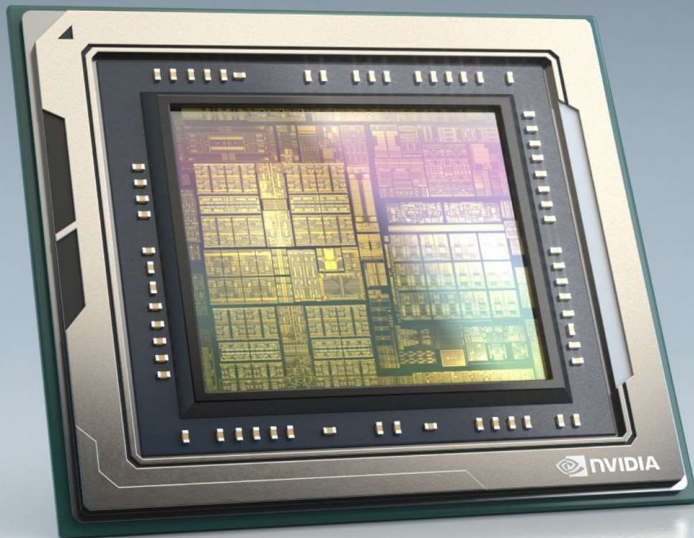


# NVIDIA - ROADMAP

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| NVIDIA ARM SoC Specification Comparison |         |                                    |                                      |
|-----------------------------------------|---------|------------------------------------|--------------------------------------|
|                                         | Orin    | Xavier                             | Parker                               |
| CPU Cores                               | ?       | 8x NVIDIA Custom ARM "Carmel"      | 2x NVIDIA Denver + 4x ARM Cortex-A57 |
| GPU Cores                               | ?       | Xavier Volta iGPU (512 CUDA Cores) | Parker Pascal iGPU (256 CUDA Cores)  |
| DL TOPS                                 | [A Ton] | 30 TOPS                            | N/A                                  |
| FP32 TFLOPS                             | ?       | 1.3 TFLOPs                         | 0.7 TFLOPs                           |
| Manufacturing Process                   | 7nm?    | TSMC 12nm FFN                      | TSMC 16nm FinFET                     |
| TDP                                     | ?       | 30W                                | 15W                                  |

# NVIDIA ORIN PRELIMINARY



sed  
es  
cks  
ock)  
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t

CPU: 12x  
Arm Cortex-A78AE  
(Hercules)  
ARMv8.2-A (64-bit)

## NVIDIA Drive PX Generation Comparison:

| Product Name       | NVIDIA Drive PX                | NVIDIA Drive PX 2                   | NVIDIA Drive Xavier                         | NVIDIA Drive Pegasus                        | NVIDIA Drive AGX Orin |
|--------------------|--------------------------------|-------------------------------------|---------------------------------------------|---------------------------------------------|-----------------------|
| SOC Name           | Tegra X1                       | Parker                              | Xavier                                      | Xavier                                      | Orin                  |
| Process Technology | 20nm SOC                       | 16nm FinFET                         | 12nm FinFET                                 | 12nm FinFET                                 | TBA                   |
| SOC Transistors    | 2 Billion (Tegra X1)           | N/A                                 | 7 Billion (Xavier)                          | 7 Billion (Xavier)                          | 17 Billion (Orin)     |
| GPU Architecture   | Maxwell (256 Core)             | Pascal (256 Core)                   | Volta (512 Core)                            | Volta (512 Core)                            | Ampere?               |
| CPU                | 16 Core ARM CPU                | 12 Core ARM CPU                     | 8 Core ARM CPU                              | 16 Core ARM CPU                             | 12 Core ARM CPU       |
| CPU Architecture   | 8x Cortex A57<br>8x Cortex A53 | 4x Denver<br>8x Cortex A57          | Carmel ARM64 8 Core CPU (8 MB L2 + 4 MB L3) | Carmel ARM64 8 Core CPU (8 MB L2 + 4 MB L3) | ARM Herclues Cores    |
| Compute DLTOPs     | N/A                            | 20 DLTOPs                           | 30 TOPs                                     | 320 TOPs                                    | 200 TOPs              |
| Total Chips        | 2 x Tegra X1                   | 2 x Tegra X2<br>2 x Pascal MXM GPUs | 1 x Xavier                                  | 2 x Volta<br>2 x Turing                     | 1 x Ampere            |
| System Memory      | LPDDR4                         | 8 GB LPDDR4 (50+ GB/s)              | 16 GB 256-bit LPDDR4                        | LPDDR4 + GDDR6                              | N/A                   |
| Graphics Memory    | N/A                            | 4 GB GDDR5 (80+ GB/s)               | 137 GB/s                                    | 1 TB/s                                      | 200 GB/s              |
| TDP                | 20W                            | 80W                                 | 30W                                         | 500W                                        | TBA                   |

| Qualcomm's Growing 64-bit SoC Lineup |                                            |                                            |                                            |
|--------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| Marketing Name                       | Snapdragon 615                             | Snapdragon 610                             | Snapdragon 410                             |
| Internal Model Number                | MSM8939                                    | MSM8936                                    | MSM8916                                    |
| <b>Manufacturing Process</b>         | 28nm LP                                    | 28nm LP                                    | 28nm LP                                    |
| <b>CPU</b>                           | 8 x ARM Cortex A53                         | 4 x ARM Cortex A53                         | 4 x ARM Cortex A53 1.2GHz+                 |
| <b>ISA</b>                           | 32/64-bit ARMv8                            | 32/64-bit ARMv8                            | 32/64-bit ARMv8                            |
| <b>GPU</b>                           | Qualcomm Adreno 405                        | Qualcomm Adreno 405                        | Qualcomm Adreno 306                        |
| <b>H.265 Decode</b>                  | Yes                                        | Yes                                        | No                                         |
| <b>Memory Interface</b>              | 1 x 64-bit LPDDR2/3                        | 1 x 64-bit LPDDR2/3                        | 1 x 64-bit LPDDR2/3                        |
| <b>Integrated Modem</b>              | 9x25 core, LTE Category 4, DC-HSPA+, DS-DA | 9x25 core, LTE Category 4, DC-HSPA+, DS-DA | 9x25 core, LTE Category 4, DC-HSPA+, DS-DA |
| <b>Integrated WiFi</b>               | Qualcomm VIVE 802.11ac                     | Qualcomm VIVE 802.11ac                     | Qualcomm VIVE 802.11ac                     |
| <b>eMMC Interface</b>                | 4.5                                        | 4.5                                        | 4.5                                        |

# COMPARISON ON CORTEX-A CORES

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[https://en.wikipedia.org/wiki/ARM\\_Cortex-A78](https://en.wikipedia.org/wiki/ARM_Cortex-A78)

## Design Philosophies



### Arm Cortex-A78

- Area and power efficiency-first design
- Designed for sustained performance within a thermal envelope
- Optimize width and depth for most efficient performance



### Arm Cortex-X1

- Performance-first design using Cortex-X Custom program engagement
- Designed for peak performance in shorter bursts
- Pushing the limits of width and depth, while maintaining max frequency capability

newest core: Cortex-X (2021)